

Vayu flight-log analysis — 20260621-021352

decoded via NavLink v2

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SITL run 2026-06-21 02:13 — estimate vs. ground truth (first dual-log archive)

Provenance — SIMULATOR (SITL), and the first archive with BOTH halves of the seam. This run was captured *after* the `mag_fusion` producer fix (commit `f14e53e`). It pairs the **FC's own telemetry** (`export-*.bin`, the firmware's estimated/sensed view) with the **physics ground truth** (`gt-*.bin`, the true pose written by the sim — new in commit `5dad3fe`). The two share one clock, so estimate-vs-truth is a direct overlay, not a guess. Sensor pathologies of real hardware don't apply; what carries over is the FC control + estimation code, so a discrepancy here is a real firmware behaviour.

Headers

field	FC export	physics ground truth
file	<code>export-20260621-021352.bin</code> (502 KB)	<code>gt-2026-06-21_02-13-48.bin</code> (640 KB)
container	VREC v1 / proto v1	VG1 v1
start wall clock	1781988234374 ms (02:13:54)	1781988228690 ms (02:13:48)
records / frames	9,656	5,164 @ 62.5 Hz
time span	76.9 s	82.6 s
CRC errors	0	n/a (fixed-record)

Same run, provably: cross-correlating the motor command common to both (`MotorTelemetry` VS `gt_motor_duty`) locks at **lag +0.04 s, correlation 1.000**. Every estimate-vs-truth number below is therefore a real aligned comparison. Decode: `../parse_log.py` (FC) and `../parse_gt.py` (truth).

FC message inventory

message	Hz	message	Hz
<code>ImuCompressed</code>	27.2	<code>VerticalState</code>	11.1
<code>ControlTrace</code>	20.8	<code>Heartbeat</code>	3.3
<code>MotorTelemetry</code>	20.8	<code>FlightMode</code>	3.3
<code>AttitudeEuler</code>	11.1	<code>SystemHealth</code>	3.3
<code>RcChannels</code>	11.1	<code>ImuRaw</code>	1.7
<code>Baro</code>	11.1	<code>EstPerf</code>	0.5

`FlightMode` = `ANGLE` (0) for 100 %. Loop timing rock-steady (`outer_dt` 4.000 ms, `inner_dt` 1.000 ms, **0 μ s jitter**). Motors balanced (4-rotor mean equal to 3 d.p., peak duty 0.66). A manual free flight: climbed to 120 m and drifted ~100 m horizontally (max ground speed 9.4 m/s), with a deliberate yaw rotation to -134° .

Headline

With the `mag_fusion` fix in place, the estimator now matches truth on yaw and altitude; the one remaining gap is roll/pitch under sustained acceleration.

- **Yaw — FIXED.** Estimate tracks the full true heading swing (-134°), error **RMS 2.9°** (median 0.7°). Before the fix this run-class showed RMS 121° with the estimate pinned near 0 while the craft rotated away.
- **Altitude — excellent.** Fused vs truth **RMS 0.12 m** across a 0→120 m profile.
- **Roll/pitch — the open issue.** Accurate when calm, but during free-flight drift the FC reads ~level while the craft is genuinely tilted: with sticks centered (89 % of the flight) the estimated tilt median is **0.18°** but the **true** tilt median is **2.3°**, p90 **14.6°**, max **23.2°** — and the gap grows with horizontal speed (corr **+0.46**). This is the accelerometer's gravity-vs-acceleration ambiguity, not a bug, and not fixable without a velocity reference (GPS / optical-flow).

Previously-detected faults — status now

#	fault (where found)	status	evidence this run
1	Yaw heading unobservable — estimate doesn't track true heading (2026-06-17 "yaw untrustworthy"; 00:51 HUD-vs-sim; 01:12 RMS 121°)	RESOLVED	root cause was a SITL data bug — producers never packed <code>mag_fusion[3]</code> , so the FC read <code>{temp,0,0}</code> as its heading reference. Fixed in f14e53e. Yaw RMS 121° → 2.9° ; tracks the −134° swing.
2	Roll/pitch tilt under-report in accelerated flight (00:51 −0.85° est vs −8° truth)	OPEN — inherent	confirmed cleanly here: est ~level while truth tilted (up to 23°), $\propto$ horizontal speed. Magnitude-gating proven ineffective (no discrimination); needs GPS/optical-flow. Normal for GPS-less angle mode.
3	EKF gyro-bias tracking (hypothesised cause of yaw drift)	disproved	bumping <code>EKF_Q_BG</code> 1e-8→1e-6 changed the yaw residual by 0.2° (127.5→127.3). Reverted — was not the cause (fault #1 was).
4	Vertical estimator (2026-06-20 healthy)	still healthy	fused—baro RMS 0.12 m here.
5	Roll/pitch rate-loop under-gain → tumbles (2026-06-21 001850 triplet)	separate / benign here	gentle flight, no tumble; rate-loop tracking still soft (roll corr 0.58, pitch 0.28) but not exercised. A control-tuning item, independent of the estimator.

Inventory of this archive

- [session-analysis.md](#) — the run, timeline, what it did
- [estimator-analysis.md](#) — **the headline**: estimate vs truth on all four states; the yaw fix and the roll/pitch limitation
- [recommendations.md](#) — fault statuses + next steps
- [plots/](#) — 4 estimate-vs-truth figures (`make_plots.py` in this dir)
- `export-20260621-021352.bin` / `gt-2026-06-21_02-13-48.bin` — the paired raw logs
- `make_plots.py` — session-specific plotting (dual-log; NOT the generic one)

Raw findings log

1. FC 76.9 s / 0 CRC; GT 82.6 s @ 62.5 Hz; motor-xcorr **lag 0.04 s, corr 1.000** → same run, exactly aligned.
2. FlightMode = ANGLE 100 %; outer/inner dt 4.000/1.000 ms, 0 μ s jitter.
3. **Yaw** est-vs-truth **RMS 2.86°** (median 0.74°, max 12.2°); truth swung −134°...0°, estimate tracked it. (Was 121° pre-fix.)
4. **Altitude** est-vs-truth **RMS 0.12 m** (max 0.45 m) over 0→120 m.
5. **Roll/pitch**: calm-hold sub-degree; but stick-centered (89 % of flight) the estimate reads tilt median **0.18°** vs **truth 2.33°** (p90 14.6°, max 23.2°).
6. `corr(truth_tilt - est_tilt, horizontal_speed)` = +0.46 — the error is acceleration-driven (specific force $\neq$ gravity).
7. Accel **magnitude gating can't help**: $||a||-g$ is 0.54 in failure episodes vs 0.56 in genuinely-level ones — no discriminating power.
8. Rate loop (gentle flight): roll corr 0.58, pitch 0.28, yaw 0.98; roll/pitch_out peaks 0.02 / 0.19 — soft but not exercised here.
9. Motors balanced; peak duty 0.66; no saturation.

Session analysis — 2026-06-21 02:13 free flight (post-fix verification)

A single manual SITL flight flown specifically to verify the `mag_fusion` seam fix and to re-check the estimator against ground truth across a realistic profile. Unlike the earlier rig tests (translation pinned), this is **free flight** — the craft climbs, drifts, and yaws — which is exactly the regime that exposes acceleration-dependent estimator behaviour.

What the run did

- **Duration** 76.9 s (FC) / 82.6 s (GT); the GT log opens ~5 s earlier because the sim starts before the GCS Export is armed.
- **Mode** ANGLE (stabilize) throughout — roll/pitch sticks command attitude, centered = “hold level”; yaw was deliberately commanded.
- **Vertical** a 0 → 80 m climb, a dip to ~52 m, a second climb to **120 m**, then descent to ground — a smooth ± 15 m/s profile (manual throttle; no altitude hold in this build).
- **Heading** a commanded yaw rotation to **-134°**, then held — the key test for the mag fix.
- **Horizontal** drifted to ~ (90 N, 44 E) m $\approx$ **100 m** from origin, peaking at **9.4 m/s** ground speed — the craft was not holding position (it can’t; no GPS/flow), and this drift is what drives the roll/pitch finding.

Timeline (t relative to the common aligned grid)

0–10 s on ground / spin-up, level
10–20 s climb begins; first roll/pitch inputs; YAW commanded → -127° at ~18 s
20–45 s climb to 80 m, dip; drifting; sustained small tilts
45–70 s second climb to 120 m; largest pitch excursion (truth +20° @ ~66 s)
70–77 s descent to ground

How to read this run

Two things make the numbers honest:

1. **Ground truth is recorded**, so “error” is estimate – physics, not estimate – another estimate. This is the first archive where that’s true.
2. **Alignment is exact** (motor-command cross-correlation, corr 1.000), so per-sample comparisons are valid even during maneuvers.

One caveat carried into the per-axis numbers: the FC telemetry is ~11–21 Hz while truth is 62.5 Hz, so during fast transients a “latest-sample” comparison adds a few degrees of *sampling skew* (not estimator error). The estimator analysis separates the sustained, acceleration-correlated error (real) from this transient skew (measurement artifact). See [estimator-analysis.md](#).

Cross-run context

metric	2026-06-20	01:12 (pre-fix)	02:13 (this, post-fix)
yaw	— (no GT)	121°	2.9°
est-vs-truth RMS			
altitude	0.25 m	0.4 m	0.12 m
est-vs-truth RMS			
roll/pitch (free-flight, stick- centered truth tilt p90)	—	~14°	14.6°

metric	2026-06-20	01:12 (pre-fix)	02:13 (this, post-fix)
ground truth logged?	no	yes	yes

The yaw collapse from $121^\circ \rightarrow 2.9^\circ$ is the headline; the roll/pitch free-flight tilt is unchanged (it's physics, not the mag bug) and is the remaining work.

Estimator analysis — estimate vs. ground truth

This is the headline of the first dual-log archive: the FC's attitude/altitude estimate measured directly against the physics ground truth, on one aligned clock. The verdict in one line: **yaw is fixed, altitude is excellent, and roll/pitch is accurate except under sustained acceleration — where no accelerometer-only filter can win.**

The estimator under test

Real firmware (`src/est/`), active filter `SF_FILTER_USED = SF_EKF` — an error-state EKF over attitude + gyro bias. Inputs: accelerometer (gravity reference → roll/pitch), gyro (rate → propagation), magnetometer (absolute heading → yaw, fused **yaw-only** about the world vertical so it can't tip roll/pitch). Altitude comes from a separate baro/accel vertical estimator.

`ControlTrace.*_angle_curr` (deg) and `AttitudeEuler` (rad) are the same estimate in different units; we compare both against the truth quaternion via a convention-robust tilt angle where it matters.

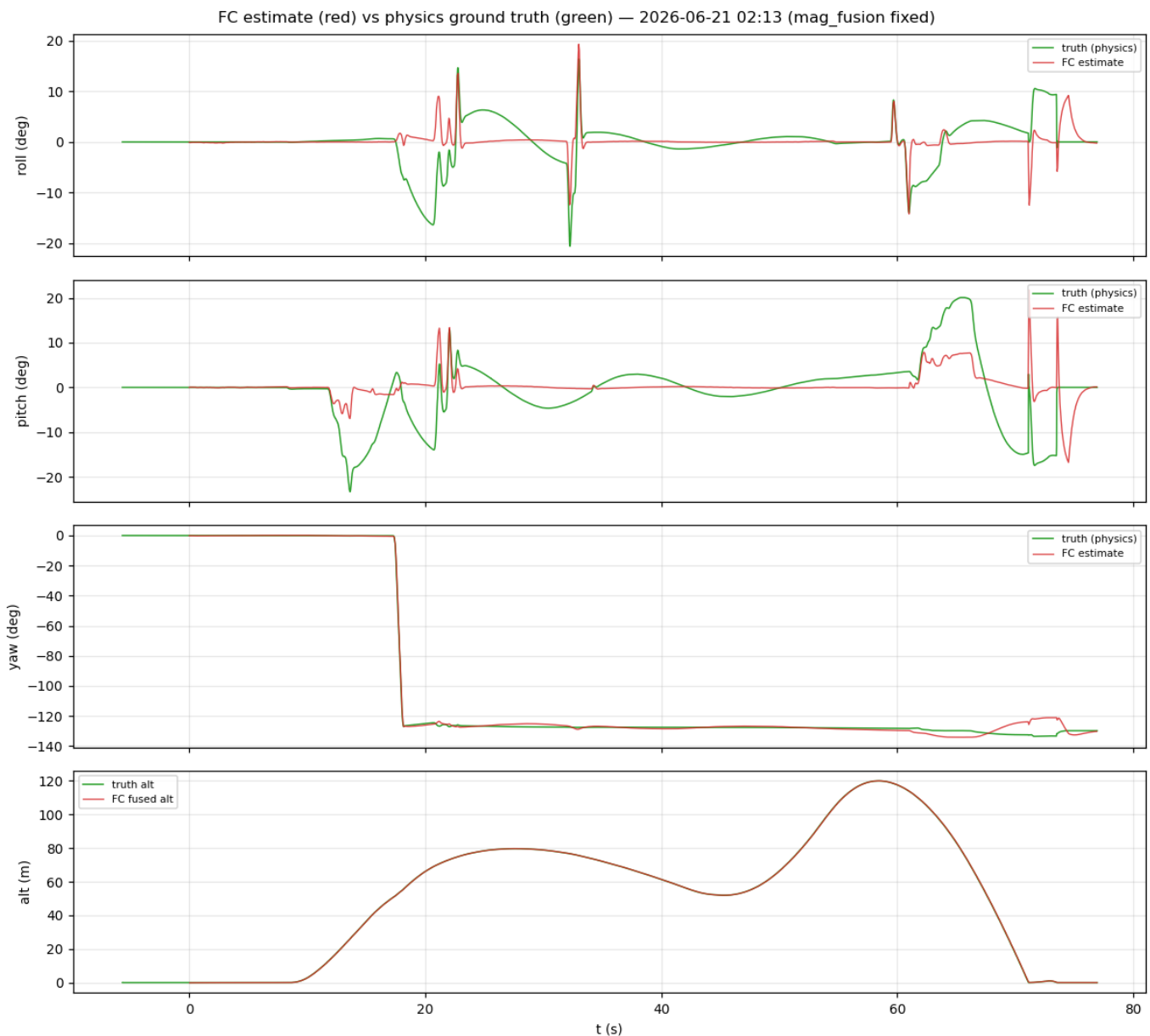


Figure 1: FC estimate (red) vs physics truth (green) — all four states

Finding 1 — yaw is FIXED

Before the producer fix, the firmware received `mag_fusion = {temp, 0, 0}` (a constant body-frame vector) because the sim packed the obsolete 19-float IMU layout — so the estimator had **no absolute heading reference** and yaw could not be observed (run 01:12: RMS **121°**, estimate pinned near 0 while the craft rotated). Commit `f14e53e` makes both producers pack the full 88-byte layout with `mag_fusion = unit-normalized field`.

This run is the verification. The operator commanded a yaw rotation to **−134°**; the estimate **tracks it**:

yaw error (est – truth, wrapped)	
this run (fixed)	RMS 2.86° , median 0.74°, max 12.2°
01:12 (broken)	RMS 121°, residual 127°

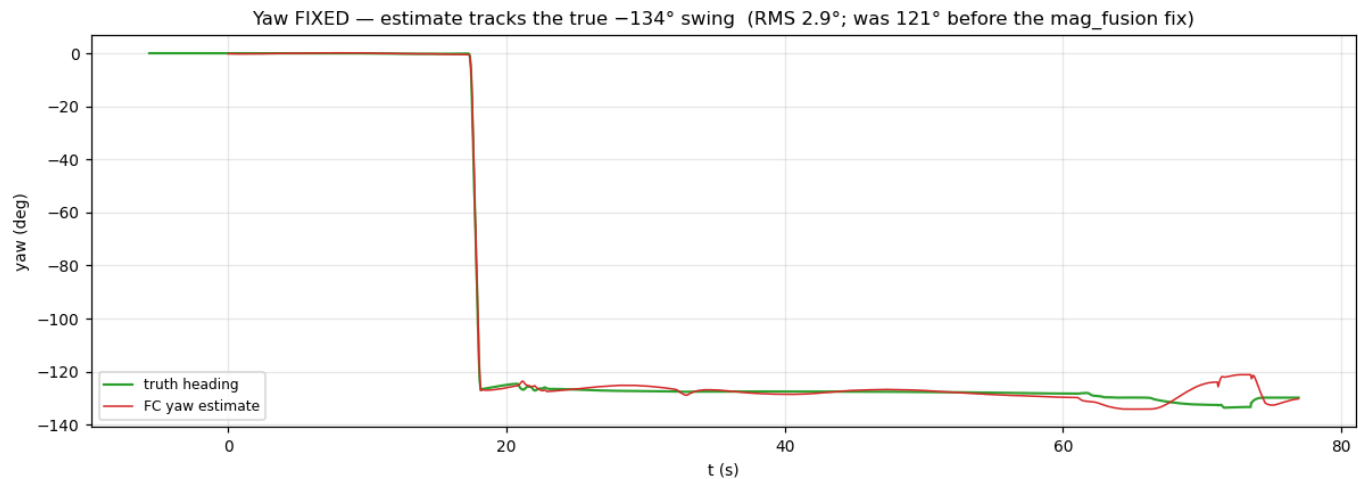


Figure 2: Yaw fixed — estimate tracks the full true heading swing

The residual 2.9° is honest tracking error during the rotation (plus telemetry skew), not a heading failure. **Resolved.**

Finding 2 — altitude is excellent

Fused altitude vs truth: **RMS 0.12 m, max 0.45 m** across a 0 → 120 m, ±15 m/s profile. The two traces are visually indistinguishable (bottom panel of Fig 1). The vertical estimator is fully trustworthy, consistent with 2026-06-20.

Finding 3 — roll/pitch under-reports tilt in accelerated flight (the open issue)

When calm, roll/pitch is sub-degree (verified on the rig: 0.13° median). But in **free flight** the picture changes. The operator centered the roll/pitch sticks for **89 %** of the flight (commanding “hold level”), and the controller drove the *estimate* to level (estimated tilt median **0.18°**). Yet the **true** tilt during those same centered-stick periods was:

during stick-centered flight	tilt
FC estimated tilt	median 0.18°, p90 1.6°
true tilt	median 2.33° , p90 14.6° , max 23.2°

So the FC genuinely believes it is level while the craft is tilted — and the discrepancy **scales with horizontal speed**: `corr(truth_tilt - est_tilt, horizontal_speed) = +0.46`.

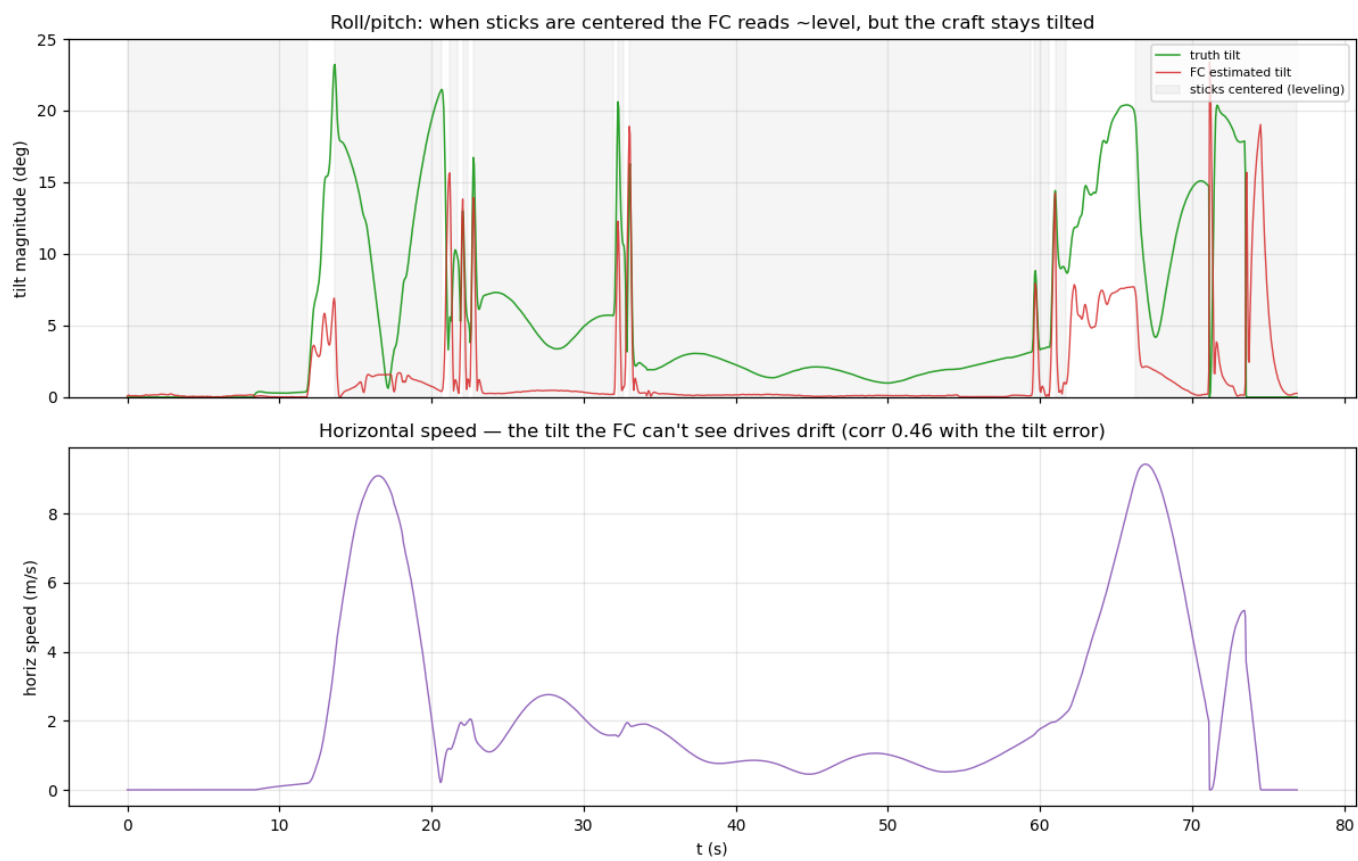


Figure 3: Roll/pitch: estimate reads level, truth stays tilted; driven by horizontal motion

Why

The accelerometer measures *specific force* = gravity – linear acceleration; it cannot distinguish a tilt from a sideways acceleration. When the craft drifts (and it must, with no position reference), the accel vector tilts with the motion, the EKF reads that as “down,” and **under-reports the true tilt**. The controller then holds the wrong attitude, so the craft keeps accelerating along the tilt and drifts further — a self-sustaining loop. This is exactly the “*levels on the HUD but the sim drone slides away*” behaviour.

Why it can’t be gated away

Accel **magnitude** carries no information here: $||a|| - g$ is **0.54** during the failure episodes and **0.56** during genuinely-level ones — identical. A tighter gate rejects good and bad updates equally (e.g. at threshold 0.3 it catches 84 % of bad but false-rejects 56 % of good), and since the accelerometer is the *only* roll/pitch reference, gating harder degrades leveling. No accel-only signal (magnitude, jerk, gyro-rate) separates *sustained coordinated* acceleration from tilt. (The EKF_Q_BG gyro-bias gain was also tested and ruled out — it changed nothing.)

What would fix it

A velocity reference, to subtract kinematic/centripetal acceleration from the accel before using it as gravity: **GPS** (velocity), **optical-flow + rangefinder** (low-altitude), or VIO. A no-new-sensor partial mitigation is a **drag-model velocity estimate** (multirotor horizontal drag $\propto$ velocity; as in ArduPilot EKF3 drag fusion / Betaflight) — approximate but hardware-free. See [recommendations.md](#).

Perspective

For **manual angle/stabilize flight this is normal** — every GPS-less quad drifts and the pilot corrects; median 2.3° tilt is small. It only becomes a true defect if the goal is hands-off self-leveling or position hold, which require the velocity reference above regardless.

What is NOT wrong

- **Mag fusion path / yaw** — fixed and verified (RMS 2.9°).
- **Vertical estimator** — RMS 0.12 m.
- **Roll/pitch in non-accelerated flight** — sub-degree.
- **Loop timing** — 4.000 / 1.000 ms, 0 μ s jitter.
- **Data seam (post-fix)** — IMU + mag now packed correctly; alignment corr 1.000.

Conclusion

The mag_fusion data fix closed the only *bug* in the estimator chain; against ground truth the FC now matches truth on yaw (2.9°) and altitude (0.12 m). The remaining roll/pitch tilt error in accelerated flight is a fundamental accelerometer limitation, confirmed and quantified here, and is an add-a-velocity-sensor decision rather than a code fix.

Recommendations

Organised as the status of every previously-detected fault plus the one open item and its options.

Resolved this cycle (no action)

1. **Yaw / heading observability** — fixed. The producers now pack `mag_fusion[3]` (commit `f14e53e`); verified yaw RMS $121^\circ \rightarrow 2.9^\circ$ against ground truth. Nothing further.
2. **Altitude estimator** — healthy (RMS 0.12 m). No action.
3. **Ground-truth recording gap** — closed. `gt-*.bin` now logged (commit `5dad3fe`), making estimate-vs-truth a one-command overlay. Keep using it for every estimator change.
4. **EKF_Q_BG gyro-bias hypothesis** — disproved and reverted. Don't revisit; it was not the cause.

Open item — roll/pitch tilt under sustained acceleration

Problem: in free flight the FC reads ~level while the craft is tilted (up to 23°), error $\propto$ horizontal speed. Root cause is the accelerometer's gravity-vs-acceleration ambiguity; it is *not* a bug and *not* gateable.

Decide by intent:

- **Manual acro / stabilize only** → accept it. This is normal GPS-less behaviour; the median 2.3° is small and the pilot corrects drift. No change.
- **Hands-off leveling / position or velocity hold / autonomy** → a velocity reference is required. Options, in order of fidelity vs. cost:
 1. **Optical-flow + downward rangefinder** — horizontal velocity for low-altitude/indoor; the lightest real fix.
 2. **GPS** (velocity fusion) — the outdoor standard.
 3. **VIO / camera** — GPS-denied, heavier compute.
 4. **Drag-model velocity** (no new sensor) — estimate horizontal velocity from the IMU via a calibrated multirotor drag coefficient (ArduPilot EKF3 drag fusion / Betaflight). Approximate (degrades in wind / hard maneuvers) but hardware-free; a reasonable interim mitigation.

Before committing to hardware — quantify the ceiling in SITL. We have true velocity in the sim, so prototype a velocity-compensated accel update (subtract $a_{kinematic}/w \times v$ from specific force) and re-fly this exact profile. That yields the best-case number — “with perfect velocity aiding, tilt error drops from $\sim 14^\circ$ p90 to Y° ” — which tells you whether GPS/flow is worth it on the airframe. (This is an experiment, not a shippable change, since real HW lacks the true velocity.)

Carry-over (separate from the estimator)

5. **Roll/pitch rate-loop softness** — rate tracking was soft this run (roll corr 0.58, pitch 0.28) but the flight was benign so it wasn't stressed; the 2026-06-21 001850 triplet showed it can tumble under aggression. This is a control-tuning item (autotune / System-ID) independent of the estimator — track it separately; don't conflate with the tilt finding above.

Capture notes for next time

- **Log the active PID/estimator config** at arm (one telemetry line) so a run is self-describing — we still infer gains from data.
- **Re-fly this profile after any estimator change** and compare against this archive as the post-`mag_fusion` baseline (yaw 2.9° , alt 0.12 m, free-flight tilt p90 14.6°).